MILITARY SPECIFICATION SHEET

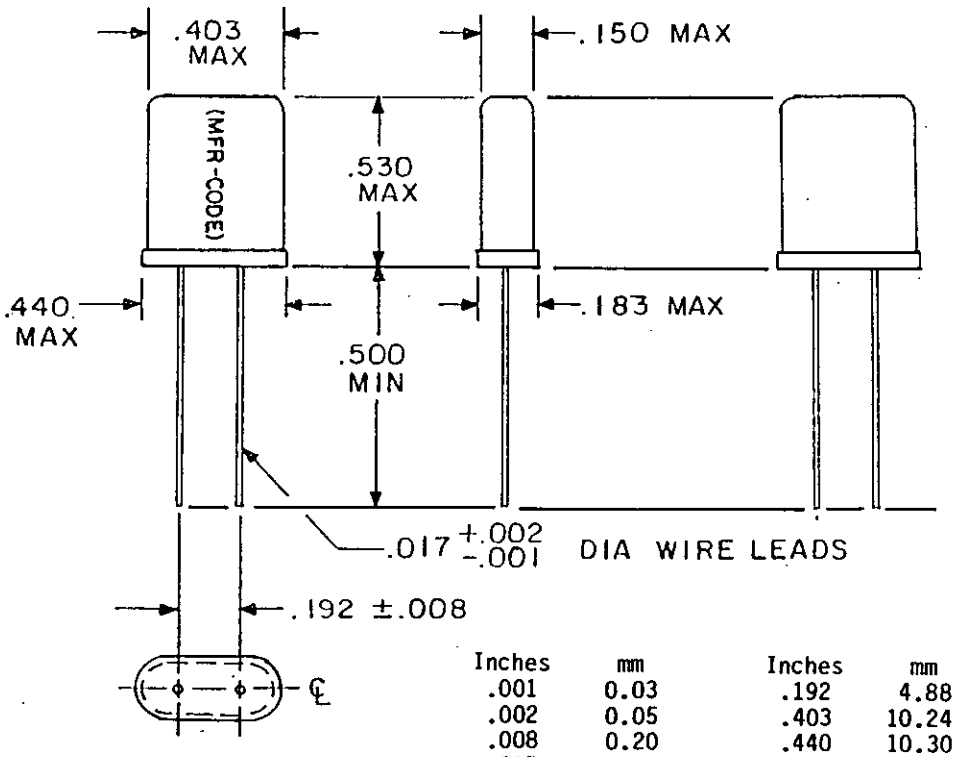
CRYSTAL UNITS, QUARTZ, PRECISION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirement for acquiring the crystals described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-49468.

Pertinent characteristics:

Frequency range:	2.4 to 50 MHz
Overtone:	Fundamental
Holder:	HC-49 (resistance weld)



Inches	mm	Inches	mm
.001	0.03	.192	4.88
.002	0.05	.403	10.24
.008	0.20	.440	10.30
.017	0.43	.500	12.70
.150	3.81	.530	13.46
.183	4.65		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Corrosion resistant metal plating of the enclosure after resistance welding is permitted.

FIGURE 1. Dimensions for configuration A.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Holder: HC-49 in accordance with MIL-H-10056/36.

Dimensions and configurations: See figure 1.

Frequency range: 2.4 to 50 MHz, inclusive.

Mode of oscillation: Fundamental overtone.

Frequency tolerance: Encoded in part number, see table III.

Frequency versus temperature: Drive level = 1.0 mA; capacitance load and environmental tolerance are encoded in part number, see tables II and III.

Frequency change permitted: $f = f_{nominal}$ ±environmental tolerance.

Resistance versus temperature: See table I.

Unwanted modes resistance: See table I.

Capacitance, shunt: 5 pF, maximum.

Capacitance, load: Encoded in part number, see table II.

Insulation resistance: 500 megohms, minimum, in accordance with MIL-C-49468.

Shock: Method 213 of MIL-STD-202, test condition D, 500g's, half-sine, 1 millisecond.

Equivalent resistance change permitted: ±10 percent.

Frequency change permitted: ±Environmental tolerance (encoded in part number, see table III).

Vibration offset: Method 204 of MIL-STD-202, test condition A.

Equivalent resistance change permitted: ±10 percent.

Frequency change permitted: ±Environmental tolerance (encoded in part number, see table III).

Moisture resistance: Method 106 of MIL-STD-202, in accordance with MIL-C-49468.

Equivalent resistance change permitted: ±10 percent.

Frequency change permitted: ±Environmental tolerance (encoded in part number, see table III).

Drive sensitivity:

Equivalent resistance change permitted: ±20 percent.

Frequency change permitted: ±2 ppm.

Condensibles:

Equivalent resistance change permitted: ±15 percent.

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Coupled modes: Polynomial - cubic (third order).

Equivalent resistance change permitted: ±10 percent

Frequency change permitted: ±3.0 ppm f < 4 MHz.
±2.0 ppm 4.0 ≤ f < 6 MHz.
±1.0 ppm 6.0 ≤ f < 10 MHz.
±0.5 ppm f ≥ 10 MHz.

Aging:

Aging temperature: 85°C ±5°C.

Equivalent resistance change permitted: ±10 percent.

Frequency change permitted (30 days): ±2.5 ppm.

Frequency change permitted per day: ±5.0 ppm.

Accelerated aging:

Equivalent resistance change permitted: ±10 percent.

Frequency change permitted: ±3 ppm.

TABLE I. Resistances.

Frequency range, inclusive	Resistance 1/ versus temperature	Resistance of 2/ unwanted modes	Drive level
MHz	Ohms	Ohms	mA
2.4 to 50 MHz	$R < R_{max}$ ($R_{max} = 1015/f^{1.6}$) where f is the frequency in MHz	$R_{unwanted} > 2*R_{max}$ ($R_{max} = 1015/f^{1.6}$) where f is the frequency in MHz	1.0 ±0.2

1/ Capacitance load and temperature range are encoded in part number, see tables II and III.

2/ Capacitance load and temperature range are encoded in part number, see table II and III. Test frequency = $f_{nominal}$ ±10 percent.

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TABLE II. Load capacitance codes.

Code	Load capacitance 1/
A	∞ pF (Series resonance)
B	100 pF
C	50 pF 50 pF 50 pF
D	30 pF
E	20 pF
F	15 pF
G	12 pF
H	8 pF

1/ The tolerance on the load capacitance should be that value which results in a frequency change not exceeding 10 percent of the frequency tolerance at the reference temperature or one percent of the nominal load capacitance whichever is smaller.

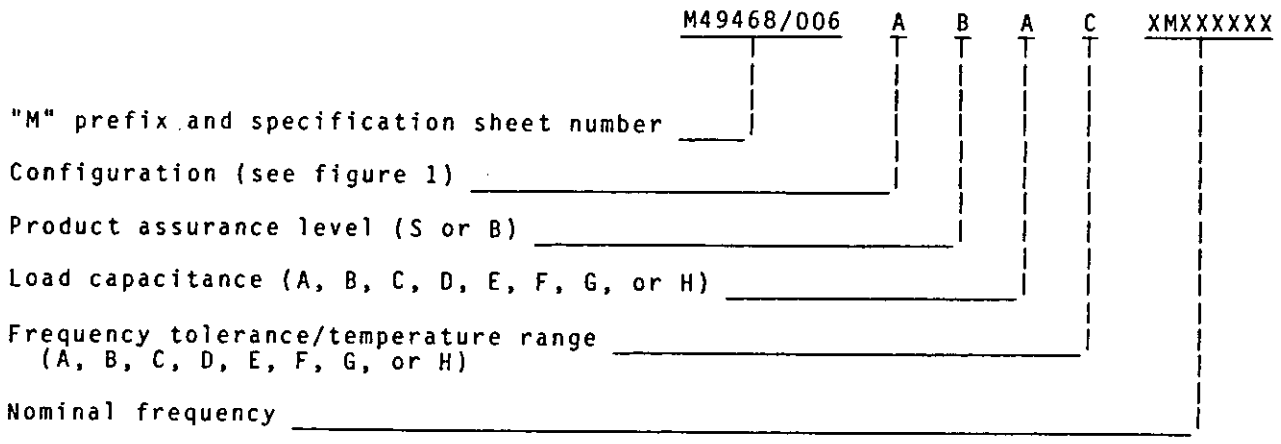
TABLE III. Temperature range and frequency tolerance codes.

Code	Temperature range	Maximum data recording interval	Maximum temperature change per time interval	Environmental tolerance	Overall tolerance
A	-55°C to +105°C	2°C	5°C/minute	±5 ppm	±50 ppm
B	-55°C to +105°C	2°C	2°C/minute	±2.5 ppm	±25 ppm
C	-55°C to -40°C	2°C	2°C/minute	±3 ppm	±30 ppm
	-40°C to +90°C	2°C	2°C/minute		±20 ppm
	+90°C to +105°C	2°C	2°C/minute		±30 ppm
D	TP = 75°C ±5°C	1°C	1°C/minute	±1 ppm	±2 ppm at TP
E	TP = 85°C ±5°C	1°C	1°C/minute	±1 ppm	±2 ppm at TP
F	TP = 95°C ±5°C	1°C	1°C/minute	±1 ppm	±2 ppm at TP
G	TP = 105°C ±5°C	1°C	1°C/minute	±1 ppm	±2 ppm at TP
H	TP = 115°C ±5°C	1°C	1°C/minute	±1 ppm	±2 ppm at TP

TP = Turnover point, see appendix D of MIL-C-49468.

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Part or Identifying Number (PIN): Consists of "M" prefix followed by specification sheet number, coded alphas, and a numeric-alpha number as shown in following example:



CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI, SM
Air Force - 99
Navy - SH
DLA - ES

User activities:
Army - ME
Navy - AS, CG, MC, OS
Air Force - 17, 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5955-0649-06)

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